MILITARY STANDARD

MIL-STD-1320-163
(NAVY)

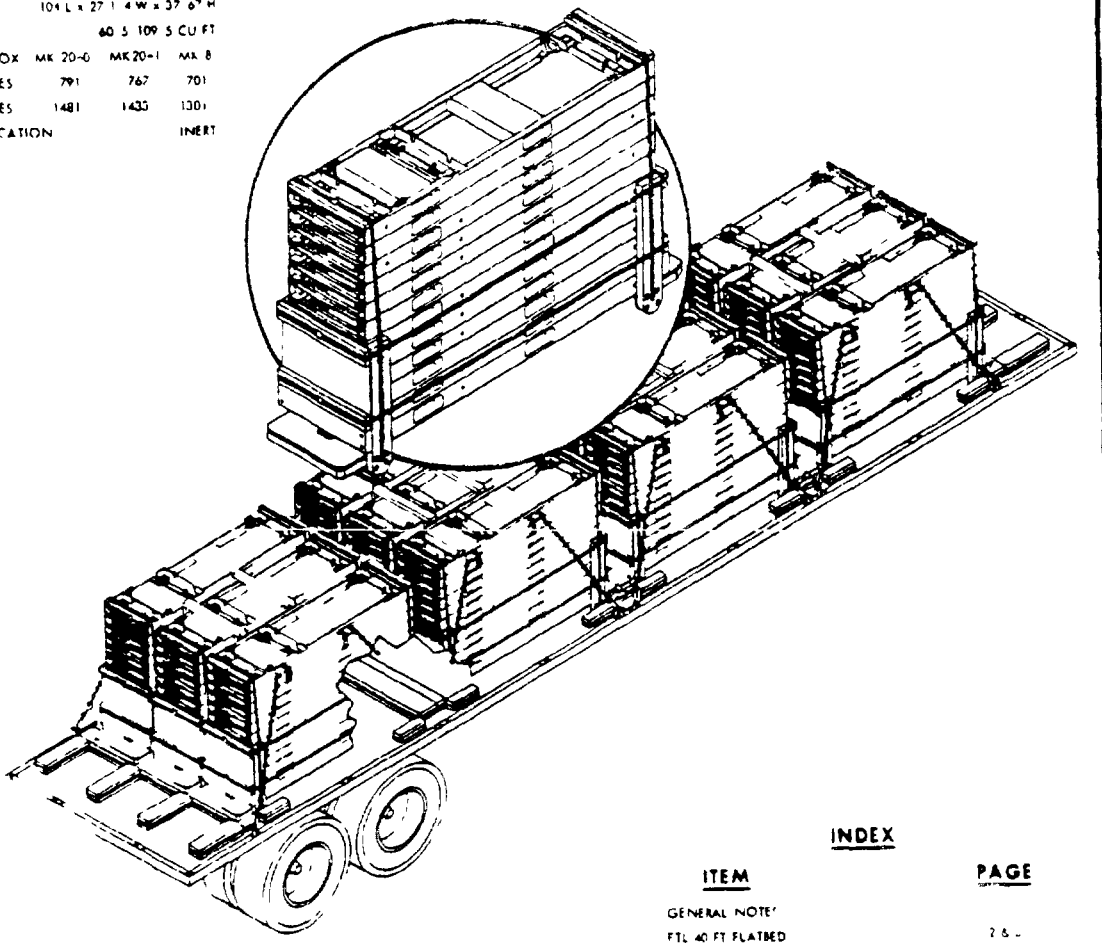
TRUCKLOADING OF HAZARDOUS MATERIALS

CRADLE, STORAGE, MK 20 MOD 0 OR 1
CRADLE, STORAGE, MK 8 MOD 0
DOMESTIC UNIT LOAD

3 JANUARY 197-

UNIT LOAD DATA

UNIT LOAD DRAWING			WR 53 856
DIMENSIONS	104 L x 27 1 4 W x 37 6? H		
CUBE			60 5 109 5 CU FT
WEIGHT LBS (APPROX	MK 20-0	MK 20-1	MK 8
6 CRADLES	791	767	701
12 CRADLES	1481	1435	1301
HAZARD CLASSIFICATION			INERT



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	
FTL 40 FT FLATBED	2 & -
LTL 40 FT FLATBED	4
FTL 40, 42 & 44 FT VAN	4 &
LTL 40 42 & 44 FT VAN	5

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

SIGNATURE — TECHNICAL DIRECTION AGENT (IDA) — DATE

SIGNATURE — SEABEES (SLOW BY DIRECTION) — DATE

ORIGINATOR — SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE NEW JERSEY

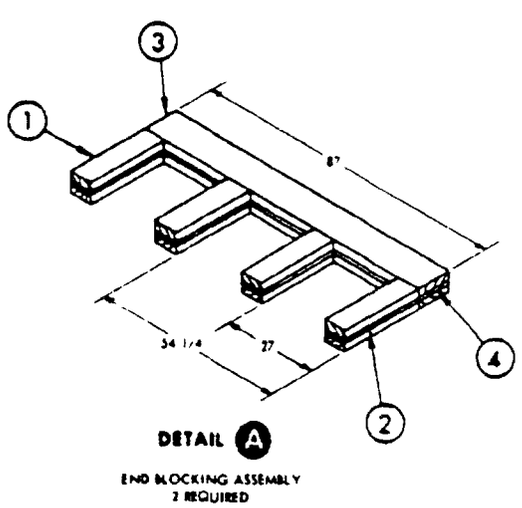
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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTION FOR TRUCKLOADING DOMESTIC UNIT LOADS OF MX 8 OR MX 20 SHIPPING AND STORAGE CRADLES. THE CRADLES SHALL BE UNITIZED IN ACCORDANCE WITH WR-50 856.
2. THE PROCEDURE AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS AND 40, 42 AND 44 FT VANS.
3. UNIT LOADS OF 12 CRADLES SHALL BE SHIPPED ON FLATBEDS ONLY. UNIT LOADS OF 6 CRADLES MAY BE SHIPPED ON FLATBEDS OR IN VANS. IF REQUIRED TO SHIP UNIT LOADS OF 6 CRADLES ON FLATBED, FOLLOW THE BASIC PROCEDURE ESTABLISHED FOR UNIT LOADS OF 12 CRADLES.
4. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
5. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER: MM-L-751; NAILS: FF-N-105, TYPE II, STYLE 15, COMMON BRIGHT; STRAPPING: QQ-S-781, TYPE I, CLASS A.
6. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT: MIL-STD-1320 (NAVY).

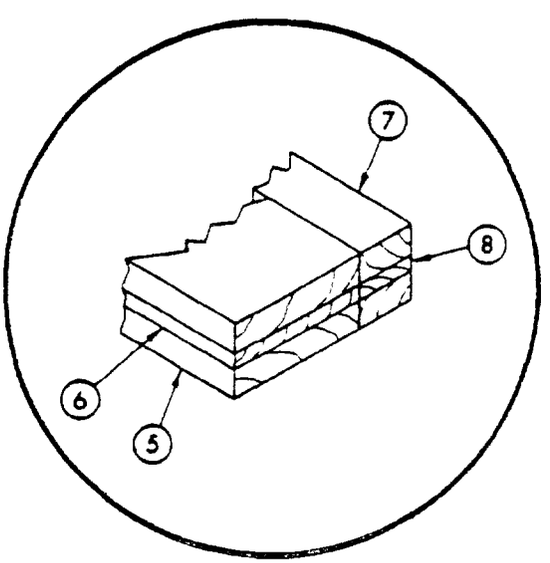
FTL 40 FT TRAILER (FLATBED)

1. SHIPMENT OF UNIT LOADS OF 6 OR 12 CRADLES EACH UNIT LOAD IS AUTHORIZED FOR FLATBED TRAILERS. DO NOT MIX UNIT LOAD OF DIFFERENT QUANTITIES OF CRADLES IN THE SAME STACK.
2. STRAP EACH STACK OF UNIT LOADS TOGETHER WITH TWO 1 1/4" GIRTH STRAPS. THE STRAPS PASS THROUGH THE FORK POCKETS OF THE TOP CRADLE IN THE UNIT LOAD, ACROSS TO THE OPPOSITE SIDE OF THE TRAILER AND RETURNS THROUGH THE THIRD LEVEL (FROM THE TOP) FORK POCKETS.
3. STEEL STRAPS ARE NOT AUTHORIXED FOR USE AS TIE-DOWNS FOR THIS TRUCKLOAD.
4. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   a. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   b. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   c. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
5. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, DOUGH'D, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.



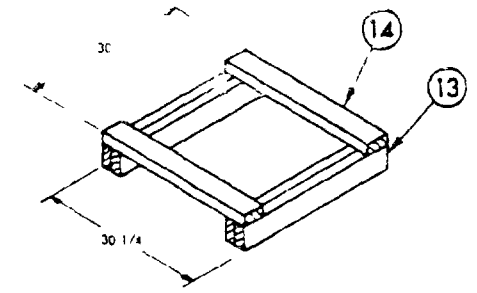
DETAIL A

END BLOCKING ASSEMBLY
2 REQUIRED

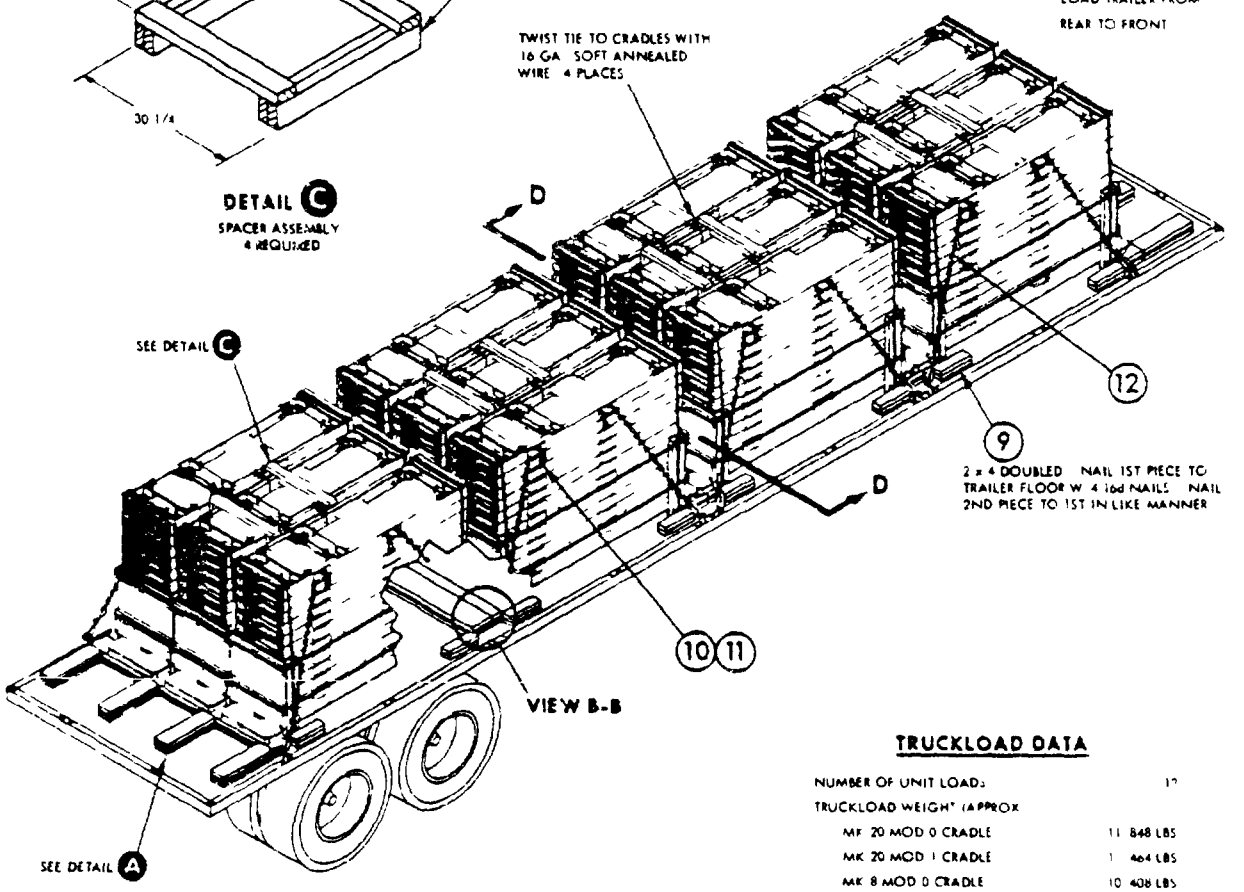


VIEW B-B

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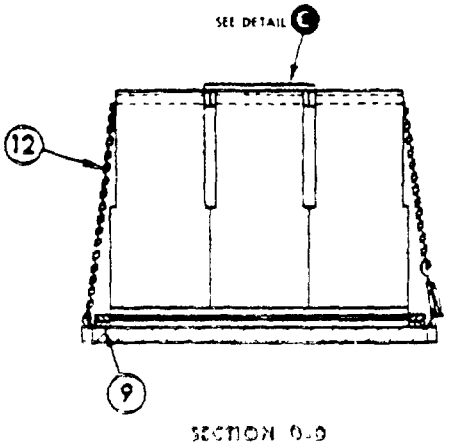


DETAIL C
SPACER ASSEMBLY
4 REQUIRED



TRUCKLOAD DATA

NUMBER OF UNIT LOADS	1?
TRUCKLOAD WEIGHT (APPROX	
MK 20 MOD 0 CRADLE	11 848 LBS
MK 20 MOD 1 CRADLE	1 464 LBS
MK 8 MOD 0 CRADLE	10 408 LBS



SECTION B-B

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
14	TIE	2 x 4 x 30 1/4	8	3	JOINT	16d
13	SPACER	2 x 4 x 30	16			
12	CHAIN & BINDER	3/8	4			
11	SEAL	FOR 4 STRAP	8			
10	STRAP	1 4x .035 x (8 Ft)				
9	SLEEPER	2 x 4 x 14	3?	SEE FIELD NOTE		
8	CROSSMEMBER	1 x 4 x 82	2	SEE		
7	CROSSMEMBER	2 x 4 x 8?	6	FLOOR THRU 1	1	16d
6	CROSSMEMBER	1 x 10 x 8?	2	SEE		
5	CROSSMEMBER	1 x 10 x 82	6	FLOOR THRU 1 & 3	16	16d
4	CROSSMEMBER	1 x 10 x 82	2	SEE 2	8	
3	CROSSMEMBER	1 x 10 x 82	4	FLOOR THRU 1	8	8d
2	BACKUP CLEAT	2 x 6 x 24	8	SEE		
1	BACKUP CLEAT	2 x 6 x 24	6	FLOOR THRU 2	6	8d

LIST OF MATERIALS AND NAILING DATA

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LTL 40 FT TRAILER (FLATBED)

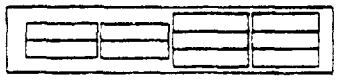
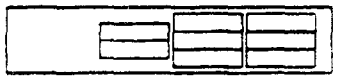
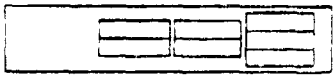
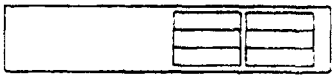
1. WHEN REQUIRED TO SHIP 6 TO 11 UNIT LOADS OF 12 CRADLES EACH SELECT THE CORRECT LOAD PATTERN FROM TABLE I BLOCK AND BRACE THE LOAD FOLLOWING THE PRINCIPLES OF THE FULL TRUCKLOAD ON PAGE 3

NOTE

WHEN REQUIRED TO SHIP 48 CRADLES OR LESS A VAN TRAILER MAY BE USED SEE INSTRUCTIONS FOR VAN TRAILERS

2. USE SWAY BRACE ASSEMBLY DETAIL F, PAGE 6 BETWEEN STACKS OF TWO UNIT LOADS TWIST TIE SPACER IN SPACER IN PLACE WITH 16 GAUGE SOFT ANNEALED IRON WIRE 4 PLACES

TABLE I

NO OF UNIT LOADS	LOAD PATTERN	FWD
11		
10		
9		
8		
7		
6		

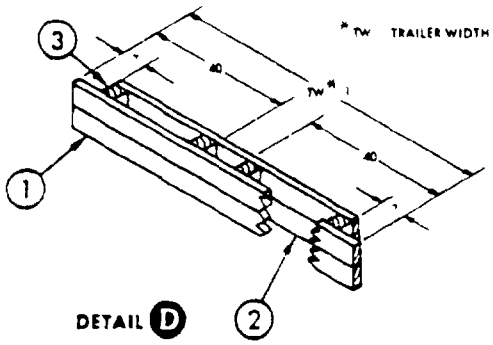
FTL 40, 42 & 44 FT TRAILERS (VANS) -6 CRADLE UNIT LOAD ONLY

1. WHEN REQUIRED TO SHIP 12 OR LESS UNIT LOADS OF 6 CRADLES EACH A VAN TYPE TRAILER MAY BE USED

WARNING

UNIT LOADS FOR VAN LOADING SHALL NOT CONTAIN MORE THAN 6 CRADLES

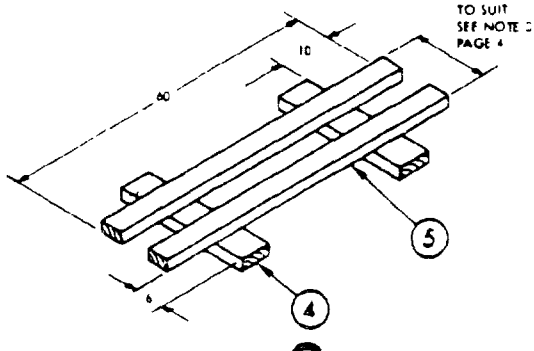
2. IF REQUIRED TO USE A TRAILER WITH A NON-NAILABLE METAL FLOOR USE THE FILLER ASSEMBLY DETAIL H PAGE 6 AT THE REAR OF THE LOAD THE ASSEMBLY SHALL FILL THE VOID BETWEEN THE UNIT LOADS AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS WHEN BLOCKING BY THIS METHOD
3. SWAY BRACE DETAIL E IS ASSEMBLED IN PLACE IN THE FORK POCKETS OF THE THIRD CRADLE FROM THE TOP OF THE UNIT LOAD THE SWAY BRACE SHALL FILL THE VOID BETWEEN THE UNIT LOADS AND CAUSE THEM TO FIT TIGHT ACROSS THE TRAILER



* TW = TRAILER WIDTH

DETAIL D

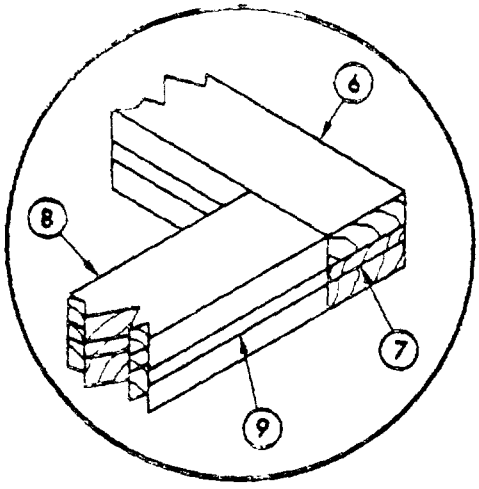
FORWARD BULKHEAD
1 REQUIRED



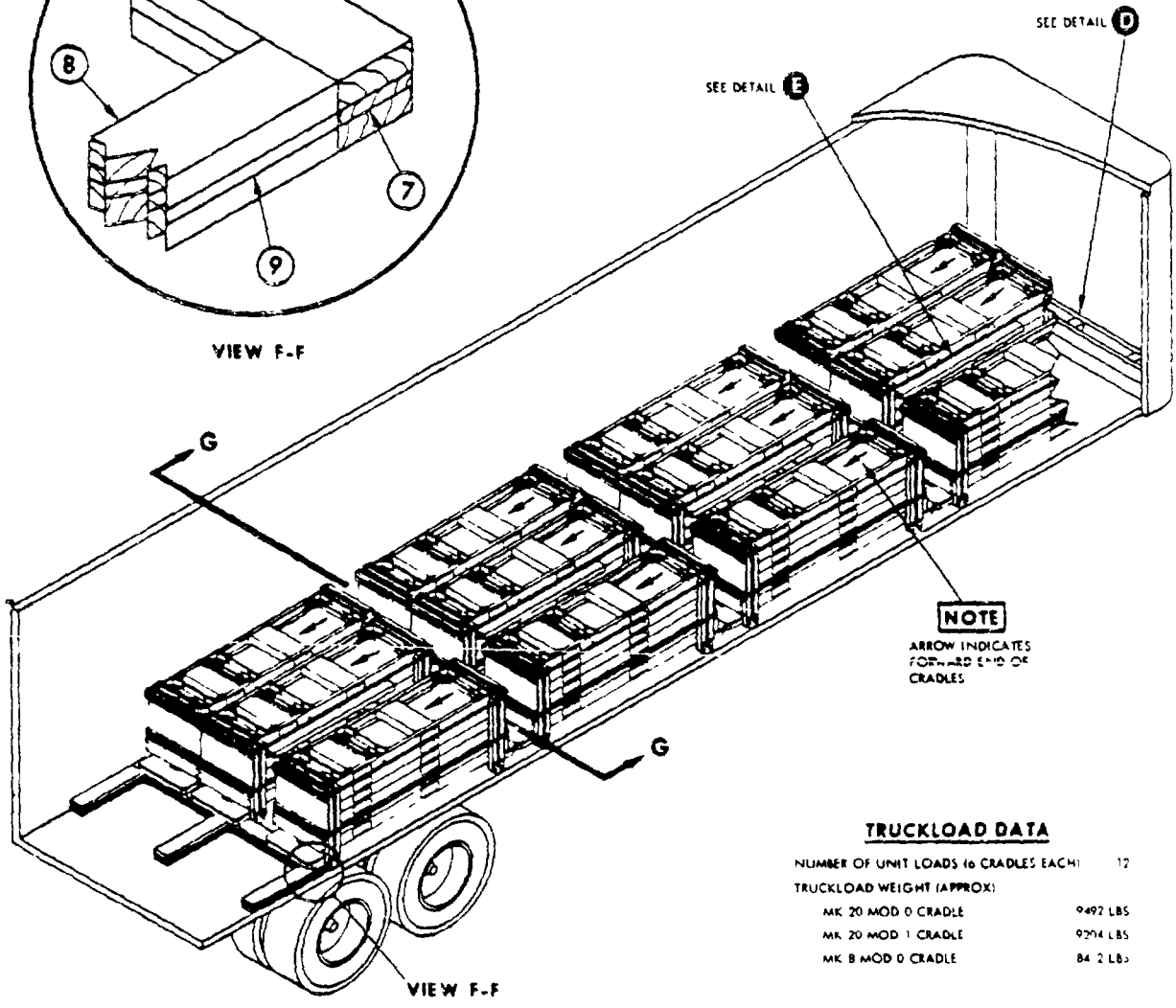
DETAIL E

SWAY BRACE
4 REQUIRED

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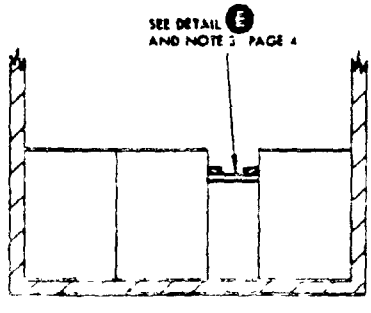
VIEW F-F



VIEW F-F

TRUCKLOAD DATA

NUMBER OF UNIT LOADS (6 CRADLES EACH)	12
TRUCKLOAD WEIGHT (APPROX)	
MK 20 MOD 0 CRADLE	9492 LBS
MK 20 MOD 1 CRADLE	9204 LBS
MK 8 MOD 0 CRADLE	84 2 LBS



SECTION G-G

*TW TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
9	CLEAT	1 x 6 x 3	3	SEE 8		-
8	CLEAT	2 x 6 x 30	6	FLOOR		10d
				THRU 9 TO 8	5	6d
7	CROSSMEMBER	1 x 10 x TW* -1		SEE 6	-	
6	CROSSMEMBER	2 x 10 x TW* -	2	FLOOR		10d
				THRU 7 TO 6	9	6d
5	STRINGER	2 x 4 x 60	8	4	2 JOINT	10d
4	SUPPORT	2 x 6 x 22	8	SEE 5		-
3	VERTICAL	4 x 4 x 11	4	SEE 1 & 2	-	
2	FWD CROSSMEMBER	2 x 6 TO SUIT		3		10d
1	AFT CROSSMEMBER	2 x 6 x TW* 1	2	-	9	10d

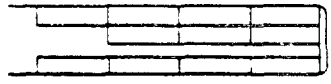
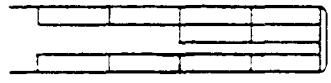
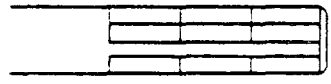
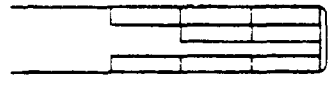
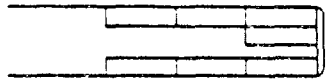
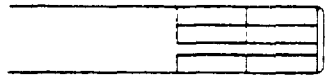
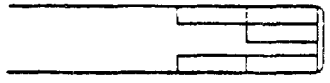
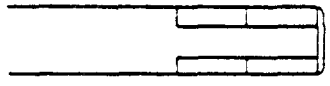
LIST OF MATERIALS AND NAILING DATA

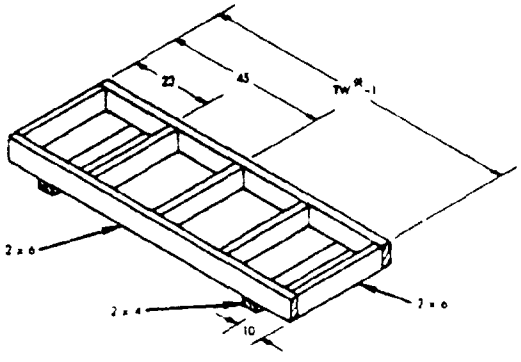
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LTL 40, 42 & 44 FT TRAILERS (VAN)

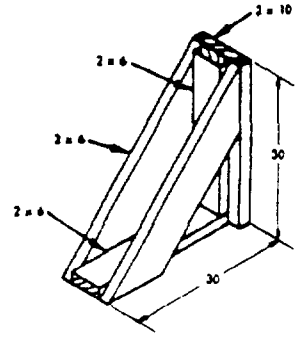
1. WHEN REQUIRED TO SHIP 11 OR LESS UNIT LOADS (6 CRADLES TO A UNIT LOAD) PICK THE CORRECT LOAD PLAN FROM TABLE II. BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGE 5 FOR THE FTL VAN LOAD
2. WHEN SHIPPING ONE UNIT LOAD USE TWO KNEE BRACES (DETAIL J) AT SIDE OF UNIT LOAD AND ONE AT THE REAR OF THE UNIT LOAD

TABLE II

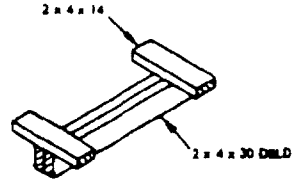
NO. OF UNIT LOADS	LOAD PATTERN	FWD
11		
10		
9		
8		
7		
6		
5		
4		
3		
2		
1		



DETAIL H
ALTERNATE REAR BLOCKING ASSEMBLY



DETAIL I
KNEE BRACE



DETAIL K
SWAY BRACE ASSEMBLY

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT NO. 8140-N267

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FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Weapons Station Earle
Naval Weapons Handling Laboratory
Colts Neck, New Jersey 07722

Code 803

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
MIL-STD-1320-163 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N 0102-014-1801 C-25254

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